

FOOTBRAWLS.GG

Product Requirements Document — v2.0 Unified

FIFA World Cup 2026 · Daily Games · Guild Wars · Raid Battles

Version	Launch Target	Status	Stack
2.0 (Unified)	June 6, 2026	Active Build — Week 4	React + Vite + Firebase + Vercel

One-line pitch: Footbrawls.gg is a daily football mini-game arcade, country guild social hub, and raid battle system built around FIFA World Cup 2026. Play games, earn XP, power your country castle, and battle rival guilds. No sign-up required.

1. Vision & Product Overview

Detail	Value
Product Name	Footbrawls.gg
Launch Target	June 6, 2026 (5 days before World Cup 2026)
Platform	Web — React + Vite, Mobile-first, PWA installable
Stack	React + Vite, Firebase Firestore, Vercel cron, API-Football
Auth	UUID + localStorage — zero sign-up friction
Monetisation	Google AdSense + Sponsorships + Footbrawls Pro (Round of 16)

Core Game Loop

- User visits → picks nickname, home country, support team
- Plays daily games → earns XP (max 200 XP per day)
- 80% of XP powers home country Castle HP / 20% powers support team Castle HP
- Castle HP determines raid strength: Fortress / Standing / Weakened
- Team wins a real WC match → guild gets Blessed (+25% XP for 24 hours)
- Team loses → guild gets Cursed (-25% XP reduction, lifted by 3 raid wins)
- Find a buddy → challenge a rival guild duo → 3-act raid battle

2. Tech Stack

Layer	Technology	Purpose
Frontend	React + Vite	All pages and all games — single codebase
Routing	React Router v6	Maps URLs to React page components
Database	Firebase Firestore	Guilds, users, chat, fixtures, predictions, raids
Auth	UUID + localStorage	No sign-up. UUID generated on first visit, mirrored to Firestore
Hosting	Vercel	Static deploy + serverless cron endpoints
External API	API-Football	Live match scores — accessed by cron ONLY, never by client
State	useState + onSnapshot	Local game state + live Firebase listeners
Share Cards	Canvas API (in-browser)	Generate result images client-side — no server cost
Push Notifications	Web Push API + Service Worker	Daily game reminder — critical for retention
Ads	Google AdSense + Rewarded Video	Apply Week 1 — approval takes 7–14 days
Moderation	JS word filter + Firestore report flags	Client-side blocklist on send — P0 feature

3. File Structure

Everything is React. No vanilla HTML/JS. All pages and all games are React components inside `src/`.

Path	Purpose
<code>public/</code>	Logo, favicon, flag images (FlagCDN)
<code>data/players.json</code>	~700 WC players (built by <code>scripts/fetchSquads.js</code>)
<code>data/fixtures.json</code>	64 WC fixtures
<code>data/questions.json</code>	Question bank for future use
<code>src/main.jsx</code>	React entry point
<code>src/App.jsx</code>	React Router — maps URLs to page components
<code>src/pages/Home.jsx</code>	Game hub + XP bar + daily game cards
<code>src/pages/Guild.jsx</code>	Guild room — chat, castle HP, curse, leaderboard
<code>src/pages/Onboarding.jsx</code>	Nickname → country → support team picker
<code>src/pages/Raid.jsx</code>	Raid lobby — buddy match, act flow, result
<code>src/pages/Leaderboard.jsx</code>	Prediction leaderboard — tournament / today / guild
<code>src/pages/games/WhoAreYa.jsx</code>	Silhouette + hints — guess the mystery player  DONE
<code>src/pages/games/Wordle.jsx</code>	Player attribute guesser
<code>src/pages/games/HigherLower.jsx</code>	Stat comparison — age / caps / goals / value
<code>src/pages/games/TransferTrail.jsx</code>	Club connection chain in fewest steps
<code>src/pages/games/MatchPredictor.jsx</code>	Predict result, scorer, scoreline  DONE
<code>src/pages/games/PenaltyNerve.jsx</code>	9-zone penalty vs AI keeper  DONE
<code>src/pages/games/DribbleGauntlet.jsx</code>	Raid-only Act 2 — zone dribble + 6-zone shot
<code>src/components/Nav.jsx</code>	Shared top nav
<code>src/components/CastleHP.jsx</code>	Live HP bar (Firestore onSnapshot)
<code>src/components/CurseBanner.jsx</code>	Curse/blessing live status
<code>src/components/ChatRoom.jsx</code>	Real-time guild chat
<code>src/components/Leaderboard.jsx</code>	Daily XP leaderboard
<code>src/components/GameCard.jsx</code>	Done/Play card for each game on Home
<code>src/components/ShareCard.jsx</code>	Canvas API share card generator
<code>src/components/RaidAct1.jsx</code>	Act 1 — solo game runner + score normaliser (0–10)
<code>src/components/RaidAct2.jsx</code>	Act 2 — Dribble Gauntlet mechanic

Path	Purpose
src/components/RaidAct3.jsx	Act 3 — Penalty Shootout with XP role assignment
src/lib/firebase.js	Firebase init — imported everywhere
src/lib/user.js	localStorage read/write — getUser() saveUser() createUser()
src/lib/xpEngine.js	awardXP() with daily cap, curse multiplier, XP split  DONE
src/lib/dailySeed.js	getDailyPlayer() — pure math, zero DB calls  DONE
src/lib/countries.js	Country list + WC teams  DONE
src/lib/matchmaking.js	Buddy system + bot fallback with ELO scaling
src/lib/scoreNorm.js	Normalise all game scores to 0–10 for Act 1 raid comparison
scripts/seedGuilds.js	Run once — creates 200 Firestore guild docs  DONE
scripts/fetchSquads.js	Pulls API-Football → data/players.json
api/cron/match-poller.js	Live score poller + curse trigger  DONE
api/cron/midnight-reset.js	Resets castle HP at midnight UTC  DONE
api/cron/resolve-predictions.js	Resolves Match Predictor after full time, awards XP  DONE
firestore.rules	 DONE
vercel.json	 DONE

4. Build Order — Stage by Stage

You are at Stage 4. Follow this exact order. Each stage unblocks the next. Do not skip ahead. Raid system is pre-launch — it ships Stage 5, before June 6.

STAGE 1 — FOUNDATION — Set up before writing a single component [DONE]

- `npm create vite@latest footbrawls -- --template react && npm install firebase react-router-dom`
- Create Firebase project → enable Firestore → copy `firebaseConfig` into `src/lib/firebase.js`
- Deploy `firestore.rules`: `firebase deploy --only firestore:rules`
- Add all env vars to Vercel dashboard (`VITE_FIREBASE_API_KEY`, `FIREBASE_SERVICE_ACCOUNT`, `API_FOOTBALL_KEY`, `CRON_SECRET`)
- Run: `node scripts/seedGuilds.js` → 200 guild documents created in Firestore instantly
- Subscribe to API-Football Basic plan → run `scripts/fetchSquads.js` → `data/players.json` created
- Push to GitHub → connect to Vercel → first successful deploy live

STAGE 2 — IDENTITY — Users need identity before games mean anything [DONE]

- Build `src/lib/user.js` — `getUser()`, `saveUser()`, `createUser()` using `localStorage`
- Build `src/pages/Onboarding.jsx` — 3 steps: nickname → home country → support team (grid of 48 WC teams)
- Wire `App.jsx` — redirect to `/onboarding` if no user found in `localStorage`
- Mirror user object to Firestore on creation (non-blocking, fire and forget)
- Increment guild `memberCount` and `supporterCount` atomically in Firestore
- Lock support team on save — no switching allowed

STAGE 3 — GAMES — Port all Crickingo games to React with football data [DONE]

- `WhoAreYa.jsx` — swap cricket players for `data/players.json` ✅ ALREADY DONE
- `Wordle.jsx` — same attribute guess logic, swap player name list
- `HigherLower.jsx` — swap cricket stats for age / caps / goals / market value
- `TransferTrail.jsx` — swap IPL teams for football clubs from `players.json`

- MatchPredictor.jsx — wire to Firebase /fixtures (written by cron job)  ALREADY DONE
- PenaltyNerve.jsx — 9-zone grid, daily AI keeper personality, 5 penalties, sudden death  ALREADY DONE
- Wire dailySeed.js into every game — getDailyPlayer(players)
- Wire awardXP(userId, source) at the end of every game result screen
- Build src/lib/scoreNorm.js — normalise all game scores to 0–10 for Act 1 raid comparison

STAGE 4 — HOME PAGE + GUILD ROOM — The hub and social layer ← YOU ARE HERE *[IN PROGRESS]*

- Build src/pages/Home.jsx — grid of all 6 daily game cards
- Build src/components/GameCard.jsx — game name, XP reward, Done or Play status from localStorage
- Build src/components/Nav.jsx — logo, daily XP bar (out of 200 cap), user chip with tier badge, midnight countdown
- Build src/pages/Guild.jsx — full guild room: left sidebar + chat + right panel (castle HP, leaderboard)
- Build src/components/CastleHP.jsx — Firebase onSnapshot on guilds/{code}, live HP progress bar
- Build src/components/CurseBanner.jsx — reads guild.currentCurse live from Firestore, red/gold banner
- Build src/components/ChatRoom.jsx — onSnapshot on /chat where guildCode == code, real-time messages
- Build src/components/Leaderboard.jsx — daily top XP earners in the guild
- Tier gating in chat — Lurkers (0 XP) read-only, Fan (50+ XP) can send, Veteran (200+) all guilds
- Profanity filter + report button — NON-NEGOTIABLE, must ship before any users see the app
- Build src/components/ShareCard.jsx — Canvas API, 1200×630px, test on REAL WhatsApp before moving on
- 48 active WC country rooms + fan guild rooms for non-WC countries (India, etc.) — all from one /chat collection

STAGE 5 — RAID SYSTEM — Pre-launch, not post-launch *[NOT STARTED]*

- Build src/lib/matchmaking.js — buddy request system, 45-sec timeout, bot fallback with ELO scaling

- Build src/pages/Raid.jsx — raid lobby: find buddy → challenge rival → act flow → result screen
- Build src/components/RaidAct1.jsx — randomly select one of 6 solo games, run it, normalise score to 0–10 via scoreNorm.js
- Build src/components/RaidAct2.jsx — Dribble Gauntlet: 3 defenders (L/C/R picks), then 6-zone goal shot, best of 5 rounds, roles swap each round
- Build src/components/RaidAct3.jsx — Penalty Shootout: cumulative XP decides Striker vs GK, same 9-zone grid as PenaltyNerve.jsx
- Wire XP role logic — higher cumulative duo XP = Striker (takes penalties), lower = GK (dives)
- Wire war records to /guilds — update castleHP on raid loss (-20% normal, -40% challenge)
- Challenge Raid mode — match day only, rival must be the IRL opponent, 2× XP stakes
- Training Mode — always available vs bots, no XP earned, no war record impact
- Bot ELO scaling — Beginner / Intermediate / Advanced / Elite based on guild rating
- End-to-end raid test on mobile — full flow from buddy request to XP payout

STAGE 6 — MATCH ENGINE — Verify live data pipeline is production-ready *[NOT STARTED]*

- Deploy api/cron/match-poller.js (already built) — runs every 2 min on Vercel, confirm in Vercel cron logs
- Test curse/blessing trigger — verify guilds.currentCurse updates correctly after a match finishes
- Test prediction resolution — verify XP is awarded with correct streak multiplier after full time
- Deploy api/cron/midnight-reset.js — verify castleHP resets at midnight UTC and timed curses expire
- Upgrade API-Football to Standard plan 1 day before June 6 — 1,500 calls/day, well within group stage needs

STAGE 7 — GROWTH — Get users and keep them coming back *[NOT STARTED]*

- Web Push notifications + Service Worker — daily game-is-live notification, critical for D7 retention
- PWA install prompt — add to home screen banner, triggers after second visit
- Apply for Google AdSense — takes 7–14 days. If not done already: do it today
- Set up rewarded reveal ad — watch ad to reveal answer in Who Are Ya / Wordle, +5 XP for user

- Set up WhatsApp Channel — daily games-are-live ping, share link in r/worldcup and football Discords

STAGE 8 — SOFT LAUNCH — Ship before the tournament starts *[NOT STARTED]*

- Target: 500 real users before June 6 (Day 0), 1,000 by Day 7 of tournament
- Post on r/worldcup, r/soccer, r/PremierLeague, football Discord servers with share card image
- Share on WhatsApp groups and Channel — each share card has game URL embedded
- Monitor Firebase console for errors, slow queries, and chat moderation issues
- Fix anything that breaks under real user load before the tournament kicks off June 11

STAGE 9 — POST-LAUNCH — Build after you have users and revenue *[NOT STARTED]*

- Footbrawls Pro at Rs. 99/month — launch at Round of 16 when users are emotionally invested
- Pro features: ad-free, unlimited reveals, gold badge, pro-only leaderboard tab
- War records public page — guild vs guild history, top raider hall of fame
- Alliance map — which non-WC countries are backing which team, visualised as a world map
- Additional games — Silhouette, Headline Year, First Touch post-launch only if DAU warrants it
- Premier League version — same codebase, new data, September 2026 Season 2

5. Data Architecture

5.1 Firestore Collections

Collection	Key Fields
guilds/{countryCode}	code, name, flag, type, castleHP, castleHPCap, currentCurse, currentBlessing, curseExpiresAt, warRecord, nextMatch, memberCount, supporterCount, raidLiftProgress
users/{uuid}	uuid, nickname, homeCountry, supportTeam, flag, totalXP, dailyXP, dailyXPDate, tier, loginStreakDays, predictionStreak, raidRecord {wins, losses, mvps}, isPro, lastActive
chat/{messageId}	guildCode (indexed), userId, nickname, flag, tier, text, timestamp, reportCount, reported
fixtures/{fixtureId}	fixtureId, homeTeam, awayTeam, kickoffAt, stage, status, homeScore, awayScore, isLive, isComplete, locksAt
predictions/{userId_fixtureId}	userId, fixtureId, predictedResult, predictedScore, resolved, resultCorrect, scoreCorrect, xpAwarded
raids/{raidId}	duoA {player1uuid, player2uuid, totalXP}, duoB {player1uuid, player2uuid, totalXP}, act1Scores, act2Result, act3Result, winner, xpAwarded, timestamp, raidType

5.2 Key Architecture Rules

- Clients NEVER call API-Football — only the cron job does, then writes results to Firebase
- Castle HP incremented atomically via FieldValue.increment() — never recalculated from scratch
- XP split: 80% home country guild / 20% support team guild — enforced in every awardXP() call
- Daily XP cap: 200 XP — enforced in xpEngine.js, resets at midnight UTC
- Chat: one flat /chat collection filtered by guildCode — one index handles all 200 guild rooms
- Daily seed: pure math, zero DB calls — seed = daysSince(tournamentStart) % playerCount

6. Games

	Game	Description	Max XP	Status
	Who Are Ya?	Silhouette + hints — guess mystery player in 8 tries	25 XP	✅ Port DONE
	Player Wordle	Guess player from attribute colour feedback	20 XP	Port
	Higher or Lower	Compare two players on age, caps, goals, market value	15 XP	Port
	Transfer Trail	Connect player A to B via shared clubs in fewest steps	20 XP	Port
	Match Predictor	Predict result, scorer, scoreline — locks 1hr before kickoff	100 XP	✅ Port DONE
	Penalty Nerve	9-zone penalty vs AI keeper with daily personality	30 XP	✅ New DONE

Removed from V1 scope: Silhouette, Headline Year, First Touch, Daily Trivia, Rapid Fire. Six focused games ship cleaner than nine mediocre ones. Add post-launch if DAU warrants it.

7. Guild & Curse System

7.1 Tier System

Tier	XP Required	Chat Access	Colour
Lurker	0 XP	Read only	#555566 grey
Fan	50+ XP	Own guild only	#60a5fa blue
Veteran	200+ XP	All guilds	#00e5a0 green
Ultra	500+ XP	All guilds	#ffd166 gold
Legend	Top 1% active users	All guilds	#a78bfa purple

7.2 Curse & Blessing System

Status	Trigger	XP Effect	Duration
Blessed	Team wins a match	+25% XP multiplier	24 hours
Cursed	Team loses a match	-25% XP multiplier	24h or 3 raid wins
Double Cursed	Loses while already cursed	-50% XP multiplier	Until 3 raid wins
Death Curse	Team knocked out	-75% XP multiplier	Rest of tournament

7.3 Castle HP

- Resets to 0 at midnight UTC daily via midnight-reset cron
- Incremented atomically on every XP award — 80% home country, 20% support team
- Hard cap: 10,000 HP per guild per day
- Above 7,000 HP = Fortress (defender advantage in raids). Below 3,000 HP = Weakened (attacker advantage).

8. Raid Battle System

8.1 Three-Act Structure

Act	Game	Format	Who Plays	Role Logic
Act 1	Random from solo pool (any of 6 games)	All 4 play — best normalised score (0–10) from each duo counts	All 4 simultaneously	Higher combined score duo wins Act 1
Act 2	Dribble Gauntlet	Best of 5 rounds — 1v1	Higher XP from Duo A vs Lower XP from Duo B (and vice versa)	XP rank within duo decides who plays
Act 3	Penalty Shootout	5 penalties, sudden death if tied	All 4 active simultaneously	Higher cumulative duo XP = Striker; Lower = GK

8.2 Dribble Gauntlet (Act 2 Full Mechanic)

Phase 1 — Beat the Defenders

- 3 defenders shown as zones: Left / Centre / Right
- Defender pre-picks which zone each of 3 defenders covers before Phase 1 starts
- Attacker picks a direction sequentially — tap, reveal, tap, reveal, tap, reveal
- Get past 2+ defenders → Phase 2 triggers. Get past 1 or less → defender wins the round.

Phase 2 — The Shot

- 6-zone goal: Top Left / Top Right / Mid Left / Mid Right / Bottom Left / Bottom Right
- Attacker picks shoot zone. Defender picks dive zone. Simultaneous reveal.
- Miss the dive → goal → attacker wins round. Correct dive → saved → defender wins round.
- Best of 5 rounds. Roles swap each round. First to 3 round wins takes Act 2.

8.3 Penalty Shootout (Act 3)

Role	Assignment	Job
Striker	Higher cumulative XP duo player	Takes penalties — picks zone from 9-zone grid. Same mechanic as solo Penalty Nerve.
GK	Lower cumulative XP duo player	Defends opponent duo shots — picks dive zone simultaneously.

Role	Assignment	Job
Simultaneous	Both duos active at same time	One duo shoots while other defends — nobody sits idle.

8.4 Raid Modes

Mode	Trigger	Win XP	Loss XP	Special Rules
Normal Raid	Any time, any opponent	+100 XP	+30 XP	Standard castle damage 20%. Always available.
Challenge Raid	Match day only vs IRL opponent	+200 XP (2×)	+30 XP (1×)	Double reward, same downside. Near 100% adoption on match days.
Training Mode	Any time vs bots	No XP	No XP	Bot ELO scales with guild rating. No war record impact.

9. XP System

Action	XP	Daily Cap
Who Are Ya? — correct	25 XP	Scales with hints used
Player Wordle — complete	20 XP	Once per day
Higher or Lower — complete	15 XP	Once per day
Transfer Trail — complete	20 XP	Once per day
Penalty Nerve — complete	30 XP	Once per day
Match prediction — correct result	30 XP	Part of 100 XP prediction cap
Match prediction — correct scorer	20 XP bonus	Part of 100 XP prediction cap
Match prediction — exact scoreline	50 XP bonus	Part of 100 XP prediction cap
Prediction streak 3+ correct	1.5× multiplier	Applied to prediction XP
Daily login streak	20 XP	Bonus 100 XP at 7-day streak
Active in chat	5 XP per hour	Max 15 XP per day
Raid win — Normal	+100 XP	Per raid
Raid win — Challenge	+200 XP	Per raid on match days
Raid loss consolation	+30 XP	Per raid
Raid MVP bonus	+50 XP	Highest scorer in any act
Share result card	+15 XP	Once per game per day
Hard daily ceiling	200 XP max	All sources combined

10. Cron & API Architecture

Core rule: Clients NEVER call API-Football. One Vercel cron fetches data and writes to Firebase. 10,000 simultaneous users = one API call per poll cycle.

Cron File	Schedule	What It Does
api/cron/match-poller.js	*/* * * * *	Poll live scores → update /fixtures → trigger curses → resolve predictions → award XP  DONE
api/cron/midnight-reset.js	0 0 * * *	Reset all guild castleHP to 0, expire timed curses and blessings  DONE
api/cron/resolve-predictions.js	On match completion	Resolve Match Predictor XP, update streak, post result card to guild rooms  DONE

11. Monetisation

Stream	When Live	Estimate
Google AdSense — banners	From launch	Rs. 1,500–2,500/day at 1,000 DAU
Rewarded reveal ads	From launch	Watch ad to reveal hint — +5 XP for user
Direct sponsorships	Week 2 of tournament	Rs. 15,000–1,00,000 per match day
Footbrawls Pro — Rs. 99/month	Round of 16	Ad-free, unlimited reveals, gold badge, pro leaderboard

Total tournament estimate: Rs. 8–9 Lakhs (conservative Rs. 3–4L, optimistic Rs. 15–20L)

12. Environment Variables

Variable Name	Where To Get It
VITE_FIREBASE_API_KEY	Firebase Console → Project Settings → Your Apps
VITE_FIREBASE_AUTH_DOMAIN	Firebase Console → Project Settings → Your Apps
VITE_FIREBASE_PROJECT_ID	Firebase Console → Project Settings → Your Apps
VITE_FIREBASE_APP_ID	Firebase Console → Project Settings → Your Apps
FIREBASE_SERVICE_ACCOUNT	Firebase Console → Service Accounts → Generate Key → paste full JSON stringified
API_FOOTBALL_KEY	api-football.com → Dashboard → Your API Key
CRON_SECRET	Generate any random string — protects cron endpoints from public access

13. What Not To Do

The most common mistakes that will eat your time without shipping anything.

- Add more games before launch — 6 games done well beats 9 games done badly. Scope is locked.
- Build raids after launch — raids are Stage 5, pre-launch. The social loop must work on Day 1.
- Call API-Football from the client — ever. Cron only. Always.
- Build complex animations — every mechanic is selection + reveal. No animation engine needed.
- Delay AdSense application — approval takes 7–14 days. Apply today if not already done.
- Over-engineer matchmaking — 45-second timeout then bot fallback. Simple is correct.
- Skip profanity filter — anonymous chat without moderation is a liability. Ships before any users.